

# THE LEDGER FRAMEWORK

## Valuation Manifesto

Governing Principles for Valuation and Profit-and-Loss — Subordinate to the Architectural Constitution and the Market Data Manifesto

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*A valuation is a projection of the record, and each mark earns its place by proving itself from the one before.*

### Abstract

A valuation is not a bare number. It is the value of a unit read at coordinates — a ledger position, a unit state, a corporate-action frame, a market-data cut, and the models the price is bound to — and the same coordinates reproduce it to the last minor unit. This manifesto states the principles that make valuation and profit-and-loss a projection of the one record and nothing beside it. Two layers, kept apart on the first page, carry the whole design: each model-priced leg *re-enters as a recorded observation*, bound to its model by lineage (MD-15); the net asset value, the profit-and-loss, and the valuation chain that assembles them are a *projection* of the record, storing nothing (C-8.2). A unit's current value is a *valuation chain*: each mark is proven from the last by a profit-and-loss explain whose residual sits below a declared bound, and an unexplained move is a broken chain, forbidden. Every mark is dispute-ready, settled by replay, extending the Market Data Manifesto's exhibit list. A corporate action is a valuation sandwich, its frame re-coordination a zero-profit identity and its near-zero residual the proof it went well. A risk report is the same valuation on shifted market data — no second system, no second code path — and a backtest is that same valuation read across a trajectory of historical or stressed world states, two strategies comparable only at identical coordinates. Each principle is an *article* derived from the seven governing principles the two parents already fixed — auditability, reproducibility, time travel, correctness, order, simulability, and the derived dispute-readiness — and rests on the clause of a parent that carries it. The manifesto is in two parts. Part A is the doctrine above — the eleven articles that govern the record and its chain. Part B, deliberately distinct, steps beneath the doctrine to state and derive one mathematical fact about the pricing function the chain stores: the price and the sensitivities a book hedges with are drawn, in general, from two different worlds, and the object that measures the gap — the *product instantaneous volatility* — is what the carry line of every explain must reconcile against. The document is rated against both parents, never the other way round; where it conflicts with either, the parent prevails.

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## PART A — Valuation Doctrine

*The record and its chain*

## 1. Valuation in the Ledger's Terms

A **valuation** is the value of one unit, together with its sensitivities, read at a fixed set of coordinates. It is never the truth about what the unit is worth; it is what a stated model says the unit is worth given stated inputs, exactly as the Market Data Manifesto holds an observation to be a fact about what a source said, not about what a thing is (MD-1). The value of the whole book is a sum of such per-unit valuations over the owned coordinate (C-4.10, C-8.2).

**Two layers, and they must never be confused.** A stored per-unit model price, or a stored greek, is a **re-entered observation**: the ledger runs no model (C-14.9), so a model is evaluated outside the fold and its output re-enters through the one door as a recorded observation (C-14.15), bound to that model by lineage (MD-15), and stale the moment an input it consumed moves (MD-8). The net asset value, the profit-and-loss, and the **valuation chain** that assembles them are of the other kind: a **projection** that computes over the record — owned quantity times price, summed over valued units — storing nothing and rebuilding from the record alone (C-8.2, C-14.11). The projection consumes the model-priced legs as *leaves* (MD-12), exactly as it consumes a directly observed stock quote: a plain observation enters the same sum. So “valuation is a pure function of ledger state” is a true statement about the projection *recipe*; “a valuation re-enters as an observation” is a true statement about each model-priced *leg*. Both hold at once, at their own layers, and every principle below follows from the pair. The distinction is not pedantry; it is where a defect is repaired. A wrong leg is a mismarked observation, re-derived with its model; a wrong total is a mis-assembled projection, rebuilt from the record. Confuse the two and you repair the wrong thing.

**Vocabulary, mapped once.** This manifesto uses the fixed names of both parents and coins no synonym for any of them (C-Auth.4). It reuses *unit*, *wallet*, *balance*, *projection*, *the immutable log*, *the market data operator*, *profit-and-loss explain*, *explain item*, and NAV; *observation*, *re-entered observation*, *frame*, *cut*, *as-of/as-at*, *complete lineage*, and *simulated path* come from the Market Data Manifesto unchanged. It coins four names, each checked against the fixed set: the **valuation record**, the **valuation chain** (always two words — the immutable log alone is hash-chained, C-4.8, and “chain” bare is reserved for it), the **profit-and-loss explain certificate**, and the **corporate-action valuation sandwich**. A scenario perturbation is a **shift** — never a “shift operator”, since *the market data operator* names one thing only, the corporate-action transform of C-9.2 (MD-13). A scenario is a *simulated path* under a recorded shift (MD-11); no “derived world” is coined.

## 2. The Articles

Each article derives from one or more of the seven governing principles and rests on a clause of a parent — the Constitution (C-*sec.n*) or the Market Data Manifesto (MD-*n*). The identifiers VM-*n* are append-only: never reused, never renumbered.

**VM-1. A valuation is a re-entered observation of a unit's price; the book value is a projection over such observations.** VM-1 The two layers of Section 1 are the first article, because every other rests on them. Where a price is read directly from the record — a listed close — it is a plain observation. Where a price is produced by a model, the ledger does not run the model (C-14.9); the model runs outside the fold and its output, together with the sensitivities computed with it, re-enters through the one door as a recorded observation bound to that model (C-14.15, MD-15) — the **valuation record**. The net asset value is then a projection: replay the log to recover the wallet's composition, multiply by the current price vector over the valued units, and store nothing (C-8.2, C-8.3). What the projection multiplies is a mixture of plain and re-entered observations; it treats them alike, as leaves (MD-12). No valuation adds a store to the ledger: it is a further fold of the same log (C-1.4).

(*Correctness, Reproducibility; C-8.2, C-14.9, C-14.15, MD-6, MD-15.*)

**VM-2. A valuation is meaningless without its coordinates; the same coordinates reproduce it bit for bit.** VM-2 A number alone is not a valuation. A valuation is fixed only by its coordinates: the ledger position it values — the composition at an *as-of* world-moment, recorded at an *as-at* cut (C-2.7, MD-4); the lifecycle state of the unit, which pricing depends on and which is read from the ledger, never from a store of its own (C-8.3); the corporate-action **frame** the value lives in (MD-13); the market-data **cut** that fixes which observations are in force (MD-6, MD-12); and the models the price is bound to (MD-15). The position’s as-at cut and the market-data cut are two coordinates, not one, and independently selectable: hold the position at yesterday’s as-at and move the market-data cut to today, and you have yesterday’s book on today’s prices — a distinct, legal query on which profit-and-loss attribution and the shifted worlds of VM-10 both rely. Given the same coordinates, any party recomputes the identical valuation, to the last minor unit (C-2.2). Reading a recorded valuation back is unconditional; *re-deriving* it from its inputs needs the retained model — the one coordinate whose retention is external to the record, a governance matter (C-14.15, §4). Because each input is pinned at its own cut and never to a mutable store, a valuation struck “mid-update” cannot arise (MD-8, MD-12): the state a valuation reads is the folded state, whole, or the valuation is not struck. A valuation carried without its coordinates is not a weaker valuation — it is not one.

(*Reproducibility, Time travel, Correctness; C-2.2, C-8.3, MD-4, MD-13.*)

**VM-3. A unit’s value is a valuation chain; each mark is proven from the last.** VM-3 The current valuation of a unit is not an isolated number but the head of a **valuation chain** — an ordered sequence of valuation records, each proven from the one before it by a profit-and-loss explain (VM-4). The chain is append-only: a re-mark is a new link — itself a re-entered observation (VM-1) — forward, never an edit of an old one (C-4.8, C-12.4), and the value the book showed at any past cut survives unrewritten beside the current one (C-12.1). The greeks that close one link seed the next: today’s exit sensitivities are tomorrow’s entry sensitivities, so the explain carries measured sensitivities at both ends of every link. Measured endpoint greeks are not a convenience: they keep the first two orders exact and squeeze any gap in Hessian coverage — the cross-greeks an implementation did not populate — into the third-order term and the residual, rather than smearing it through delta and vega, where it would corrupt the very hedge the greeks are supposed to size. The chain is the valuation layer’s instance of complete lineage (MD-6); it reaches back through the price vector and the operator-adjusted inputs to the leaf observations. A unit may carry more than one valuation chain at once — one per re-marking convention, cadence, or desk — and they do not compete: each decomposes the one change in net asset value (C-8.4) and localises to its own recorded convention coordinate (VM-5).

(*Order, Auditability; C-4.8, C-12.1, C-12.4, MD-6.*)

**VM-4. The profit-and-loss explain is the certificate, and it states honestly what is exact.** VM-4 Each link carries a **profit-and-loss explain certificate**: a decomposition of the value change between two marks, read the way a desk reads it. Carry first — the value the position sheds or earns as a day passes at unchanged market. Then the delta line — the move read against the entry delta. Then the gamma-and-theta balance, the core of the daily explain: convexity earns on the move, decay pays for it, and a delta-hedged position turns a profit only once the move clears its breakeven, the point where the day’s gamma gain just covers the day’s theta. That balance closes at exactly one volatility — the *product instantaneous volatility* of Part B (PE-4). The carry line therefore carries a declared, recorded term naming *which* of the volatilities in play it was reconciled against — the model’s, the raw implied-surface, or the product instantaneous volatility. That term is a governance lever — it decides whether the Feynman–Kac discrepancy (PE-2) sits in the carry line or lands in the residual — so a change to it is an auditable, reviewable event, exactly as the attribution convention (VM-5) and the

certification bound (VM-6) are. Then vega, bucket by bucket — by name, by expiry, by strike, never a single number — with the cross-greeks vanna and volga among the higher-order terms. Then the lifecycle and state changes the link crossed. And last, one explicit residual (VM-6). Two forms of the market lines exist and the certificate must not conflate them. The *ex-ante* estimate — the reading available *before* the move, computed from the entry greeks alone — is  $\Delta \cdot \delta + \frac{1}{2} \delta^T \Gamma \delta$ , the delta-and-gamma a desk pre-computes, and carries a *deterministic, signed* gap to the reprice equal to the variation of gamma over the move. The *exact* decomposition uses the greeks at *both* ends: the value change is the move times the average of the entry and exit deltas, less a one-twelfth correction for the gamma that moved (a both-endpoint quadrature correction), plus a residual  $R$  that is identically zero only for a book cubic in its risk factors. Using entry greeks in place of both-end greeks “is not a smaller model; it is an inconsistent one”, so the certificate names which form it used. The residual behaves well only under conditions: for a value smooth to fourth order,  $R = O(\|\delta\|^4)$  — doubling the move multiplies its leading part by sixteen; at a digital, a barrier, an autocallable, any payoff that is not smooth,  $R = O(1)$ , the size of the discontinuity, the instant a move straddles the feature; and the calendar direction is never covered by that bound — a day of carry near expiry lands in  $R$  at full size. The certificate therefore *never* promises a universally bounded residual. The total the certificate explains is path-independent: it is exactly the change in net asset value (C-8.4). Only the *decomposition* depends on the marks chosen along the way and on the convention (VM-5) — the identity that the parts sum to  $\Delta \text{NAV}$  holds on every path; the split into parts does not, and no second running total of realised profit-and-loss is kept beside NAV to disagree with it (C-8.4). Because the residual is the balancing term — the value change less every attributed line — the certificate reconciles to the change in net asset value by construction, not by coincidence: it is a projection of that one number, and no split it prints can make it disagree with the fold. Attribution decomposes  $\Delta \text{NAV}$ ; it never recomputes it. This explain *composes* with the Constitution’s reordering explain: there is one profit-and-loss explain, and a reordering-refold line (C-12.6) and a market-factor line here are two disjoint kinds under one roof, each attributed to exactly one cause. The certificate extends the taxonomy; it does not re-coin the term.

(Correctness, Auditability, Order; C-8.4, C-12.6, MD-5.)

**VM-5. Attribution is convention-dependent; the convention is a declared, recorded term.** VM-5 How a move splits across the greeks is not unique. Where a risk factor is not a market map of its driver — an implied basket correlation is the canonical case — the split between the spot channel and the vol channel depends on which quantity the desk holds invariant as the others move, and two conventions that reproduce today’s marks exactly differ in tomorrow’s attribution: same marks today, different *attributed* profit-and-loss tomorrow. That invariant is a declared, recorded term of the valuation (MD-6). Two consequences follow. The same convention reproduces the same attribution, bit for bit, a deterministic function of the record. And a dispute about attribution is a dispute about convention: replay *localises* it to the differing term (MD-14) but does not adjudicate which convention is right, which is model choice and out of scope (MD-9, MD-15). The manifesto therefore never speaks of *the* attribution as unique, nor of a shadow greek — the value routed through the held-invariant factor — as a property of the market: under the engine default that holds the at-the-money volatility constant, every spot-channel shadow number is identically zero, and must be flagged as computed under that convention, never quoted as a market fact.

(Correctness, Reproducibility; MD-6, MD-9, MD-14, MD-15.)

**VM-6. “Explained” means the residual sits below a declared bound; a breach is a recorded fact, never absorbed.** VM-6 A move is *explained* when its residual  $R$  falls inside a declared tolerance. The tolerance is a declared term, not a universal constant: a valuation is certified only when each instrument’s residual is within its bound and the aggregate is within the

aggregate bound, and those bounds are recorded in the model configuration, mapping between price space and vol space through vega. These bounds roll up: the per-unit residuals aggregate to a book-level residual against the aggregate bound, where cross-unit and correlation profit-and-loss — routed under the declared convention (VM-5) — must also come to rest. The residual is itself a re-entered observation — a recorded diagnostic, stale if its inputs later move, never a silent pass (MD-9, MD-15). One structural source of it is named in Part B: a carry line that books the model theta against the gamma the book carries but reconciles it at a volatility other than the product instantaneous volatility (PE-4) leaves the Feynman–Kac discrepancy (PE-2) sitting in  $R$ . Which volatility reconciles the line is a declared choice (VM-4); making it silently — mixing the two worlds without declaring — is exactly the unexplained profit-and-loss this article refuses to pass. It is a sensor, not a rounding error: a residual that is small, mean-zero, and regime-independent certifies the bookkeeping is a control variate — a correction whose smallness can be trusted —; a residual that is structurally large, sign-biased, or wakes up in stress reports that the honest attribution of this position is a scenario, not a greek, and demands re-mark or model review. Two things the bound must survive. First, the anti-plug: that the certificate reconciles to the change in net asset value proves *nothing* — the residual is the balancing term (VM-4), so the parts sum to  $\Delta NAV$  whether the move is explained or not; only the residual *below its declared bound* certifies. Reconciliation is automatic; explanation is earned. Second, the bound’s own adequacy is a governance assumption, named here as the parent named the identifier grain and the delivery frame: a bound set too loose makes every move “explained,” fires no breach, and empties “unexplained profit-and-loss is forbidden” (VM-7) in silence. The guard is that the residual is recorded whatever the bound — its sign and regime behaviour, not its level alone, are inspectable after the fact — and that the bound’s calibration is itself a recorded, reviewable event, so a loosened bound is an auditable change, never a silent one. A breached bound is booked — a named line in the profit-and-loss explain (MD-5) — and acted on. A bookkeeping that announces when to stop believing it is worth more than one that is elegant while it lasts.

(*Correctness, Auditability; C-2.4, C-13.2, MD-9, MD-15.*)

**VM-7. Unexplained profit-and-loss is a broken chain, and a broken chain is forbidden.** VM-7

A link whose residual exceeds its declared bound, or whose leaf inputs have moved under it, is not a link: it is a broken valuation state, and a broken state cannot hide (MD-8). Two mechanisms keep it visible. A projection over a corrected leaf recomputes on the next read, so a stale *projection* cannot arise. A re-entered valuation record that consumed a corrected or superseded input — a wrong price, a restated corporate action behind its frame — cannot re-run itself (C-14.9); its lineage marks it stale and it stands as a flagged open item for re-derivation (MD-8, MD-10), read through a projection that carries the staleness flag, never as the raw stored number. Staleness propagates forward along the chain: a superseded link flags stale every downstream link whose certificate chained from it, and each re-proves *forward from the superseding predecessor* — never editing the old proof — so a certificate can never keep attesting a transition from a mark that no longer heads its segment (MD-6, MD-8). The as-known branch survives beside the corrected one (C-12.1), so the chain resolves into the two honest answers rather than silently forking. Nothing in the chain is repaired by overwriting; a correction repairs forward, and where money already moved it moves back only under authorised compensation (C-12.4). What is forbidden is the *silent* unexplained move: an unexplained profit-and-loss is a defect the record must surface, not a number the book may carry.

(*Correctness; C-1.5, C-14.9, MD-8, MD-10.*)

**VM-8. Every valuation is dispute-ready.** The current valuation of every unit is dispute- VM-8  
ready: its chain, the certificates along it, and its coordinates suffice to reproduce and defend it, bit for bit, against any counterparty — a valuation dispute and a collateral dispute alike. This is the Market Data Manifesto’s dispute-readiness (MD-14) reached at the valuation layer, and it

*extends* that manifesto’s exhibit list: to the observation, its provenance, the cut, the frame, and the bound model, valuation adds the **valuation chain** and its **certificates**, exhibited as further rows of the one replay, never a second replay doctrine. Its bound is exact and inherited: replay settles whether the mark is the faithful, deterministic consequence of the inputs the record held at its stated coordinates, the as-at pinned to the mark’s own cut so the number that stood then is reproduced, not a later one. It does not settle a two-sided economic dispute — a counterparty’s mark, from its own surface and convention, replays bit for bit too — but exhibiting the chain and the convention *localises* the disagreement to the differing input, frame, model, or convention, which is the value replay delivers (VM-5, MD-14). A collateral dispute is bounded the same way: replay settles the *mark* inside the call, never the collateral-agreement terms — haircut, eligibility, netting — which are model choice and out of scope (C-Scope.11).

(*Dispute-readiness (Auditability + Reproducibility); C-2.1, C-2.2, MD-14, MD-15.*)

**VM-9. A corporate action is a valuation sandwich; its frame re-coordination is not profit-and-loss.** Every corporate action that changes a unit’s state triggers a **valuation sandwich**: a valuation struck immediately before the transition and another immediately after, with a full profit-and-loss explain across the pair. Across the sandwich the state and the coordinate frame change discontinuously, so the smooth explain of VM-4 does not apply — there is no smooth move to expand along. The only admissible explanation of the jump is the market data operator (C-9.2, MD-13): it transports the pre-frame mark into the post-frame, acting on leaf observations, with derived objects recomputed from operator-adjusted inputs, never scalar-transported. The frame re-coordination itself is a *zero-profit identity*: the same value in new coordinates, the consistency-of-reference that forbids multiplying a new share count by a stale price (C-11.3). Only genuine economics leaving or entering the position — a special dividend paid away — is profit-and-loss, and it earns its own explain line; there is no “corporate action profit-and-loss” beyond it. That the sandwich residual is  $\approx 0$  is the proof the action went well, and  $\approx$  means exactly three declared tolerances and no more: the minor-unit rounding applied once at read (C-4.6), the declared convention’s own rounding, and the calibration tolerance of any re-derived surface or curve (VM-6). A residual materially above these is not noise — it says the operator or the delivery frame was mis-declared, which replay localises (MD-14) and MD-8 and MD-10 handle as a superseding event demanding re-derivation. The operator exists only once the action’s terms are *resolved*, not merely announced; before resolution the frame is provisional and the sandwich is struck as such (MD-13). A third timing case is a resolved action that reaches the door *late*, after valuations were already chained across its ex-date. This is a reordering, not a lineage supersession: the intervening links never carried the action in their lineage, so they are flagged stale by the reordering trigger (C-2.7, MD-5), not by the frame-supersession path of VM-7. The action takes its place at the ex-date, its sandwich is struck *retroactively* there, and the spurious market-move line the chain had booked — a two-for-one split reaching the door three days late reads as a fifty-percent crash until it arrives — is reclassified as the zero-profit frame re-coordination, the reordering carried as its own line in the profit-and-loss explain (C-12.6). The loss is never left standing.

(*Order, Correctness, Time travel; C-9.2, C-9.3, C-11.3, MD-13, MD-5.*)

**VM-10. Risk is valuation in simulated worlds.** A risk report is not a separate system and not a second code path: it is the same per-unit valuation recipe of VM-1 applied to *shifted* market data. The Constitution already fixes the doctrine — “we live in a simulation: production is simply the one path whose events happened to be real” (C-2.8) — and the Market Data Manifesto that simulated market data is real market data under a different seed (MD-11). A scenario is a simulated path in which one or more observations are moved by a recorded **shift**; a greek is a sensitivity read across the base and shifted paths. Shifts *compose* — spot up twenty percent, volatility down, in any combination — each composition yielding a new vector of valuations

and greeks by re-application of the one recipe. Base and simulated worlds share their units, their models, and the market data operator; only the market-data state differs. And the gap between the greek approximation and a full revaluation is itself a computable, recordable fact — the ex-ante estimation gap of VM-4: the deterministic, signed gamma-variation term *plus* the residual  $R$  (VM-6), small for a smooth book and  $O(1)$  where the payoff is not — so a sound system computes both and records their difference, on the same terms as a production day, never choosing silently between them. The shift is recorded, exactly as a seed is (MD-11): it is the single non-record input of the scenario, so the path replays. But a path that recorded only its shift could not replay — every model-priced valuation along it re-enters as an observation like any other (VM-1), into the simulated path’s *own* record, never as a link in the real unit’s valuation chain — so a stress run is a derived object with full lineage, dispute-ready on the same terms as production (VM-8). And because risk and the books run the one recipe, the risk report and the official profit-and-loss cannot disagree by construction: the classic risk-versus-books reconciliation break — two systems, two code paths, two numbers — has no way to arise. The doctrine reaches one step further, and time travel, reproducibility, and simulability already imply it: let the recorded shift be the identity — what actually happened — over a base taken from a past cut (C-12.1), and the simulated path is a stretch of the real history; a valuation run across a trajectory of such cuts is a *backtest*. One doctrine then covers the risk report, the scenario, and the backtest — a valuation on a derived world, whose shifts in the backtest are what actually happened — and VM-11 develops it.

(*Simulability, Reproducibility, Time travel; C-2.8, C-12.1, MD-11, C-14.11.*)

**VM-11. A backtest is a strategy unit run through a trajectory; its output is an ordinary valuation chain.** A strategy is already a unit, and the Constitution says so (C-10.2): a deterministic strategy is a smart contract whose rulebook is the terms of a (non-valued) strategy unit — the rulebook holds no value, its portfolio does — carried in a wallet in a virtual ledger, each rebalancing a recorded transaction triggered by stamped observations. This article specialises that, restating none of it. The strategy’s ProductTerms declare the hedge instruments, the rebalancing rules, and the triggers (C-7.2); its behaviour is declared terms, not hidden code (C-2.6), and its contract a pure function of those terms, the recorded observations, and the visible state (C-6.4) — so the same rulebook, run twice on the same record, emits the same transactions, its contract firing on the trajectory’s events as a listed option’s fires on a settlement print (C-6.1).

A **backtest** is that strategy unit run through a *trajectory* of world states — each the market-data state at one historical cut, the observations then in force in their frame — the served or stressed history the Market Data Manifesto makes replayable, gap-free, and frame-correct through the horizon (MD-4, MD-11, MD-13). Its output is an *ordinary* valuation chain (VM-3): a valuation at every step (VM-1), each proven from the last by a profit-and-loss explain certificate with entry and exit greeks (VM-4), transitively repaired on a corrected input (VM-7), sandwiched across every corporate action in the span (VM-9). It is the same pure function of ledger state, the same certificates, and the same proofs as a production chain — nothing about a backtest is a second kind of valuation. The backtest chain lives in its own path’s record, never the real unit’s chain (VM-10); where its coordinates coincide with production — the same strategy over the real history — its chain *equals* the production chain by determinism (C-2.2), reached by recomputation, never a second stored copy that could drift.

A backtest reports two numbers, each a function of the chain and its declared terms. **Absolute performance** is  $A(C) = (V_N - V_0) + \sum_k F_k$ , over the chain’s marks  $V_k$  and its recorded flow lines  $F_k$  (dividends and coupons in, financing out): by the certificate’s reconciliation to  $\Delta NAV$  (VM-4, C-8.4) it telescopes to the endpoints and the flows, touches no attribution split, and is therefore *convention-independent* — the path-independent total the certificate exists to reconcile to. **Profit-and-loss volatility over the life** is  $\Sigma(C; \pi, D, \nu) = \nu D(\Delta V_1, \dots, \Delta V_N)$ , the dispersion of

the recorded per-step increments  $\Delta V_k$  under a declared dispersion rule  $D$  (say the standard deviation of the increments — the manifesto fixing none) and a declared normalisation  $\nu$  (the annualisation), over the trajectory’s own partition  $\pi$ . At principle level  $\Sigma$  is deterministic given the chain and  $(\pi, D, \nu)$  — a value read off the record, not a statistical estimate — and  $\pi, D, \nu$  are *declared, recorded terms* carrying the same auditable-change discipline as the attribution convention (VM-5) and the certification bound (VM-6); the partition is a coordinate, not a free choice, since resampling changes the increments and hence  $\Sigma$ .

Two strategies compare *validly* only at identical coordinates: the same observations at the same cuts, in the same frames, with the same bound models, over the same trajectory, and the same declared terms — differing in nothing but the strategy unit under test. That identity is a decidable check on the record: each chain carries complete lineage (VM-3, MD-6), and the two lineage sets are compared. Differ in anything but the strategy unit and the comparison measures two worlds, not two strategies; it is *invalid by definition* — the record yields no valid comparison, the exact analogue of VM-2, a comparison without shared coordinates being no comparison at all. Where validity holds, the comparison is a projection over the two chains, dispute-ready by replay (VM-8): recompute both chains and both functionals from the record, reproduce each bit for bit, and localise any divergence to a differing coordinate or term — never adjudicating which strategy is better, which is model choice and out of scope.

One bias the structure cannot refuse: look-ahead is impossible by construction (MD-4), but *survivorship* is not. The instrument universe the strategy ranges over is a declared term of the strategy unit — resolved as-known (the units in force at each historical cut, delisted names included, all on the append-only record) it is unbiased; declared as today’s survivors it runs a biased backtest the structure cannot detect. The record makes that declaration auditable, on the chain’s coordinates, but cannot police its wisdom: “look-ahead cannot arise” is not “bias cannot arise.”

That a backtest is no harder *to specify* for a sophisticated exotic than for a vanilla is not asserted but forced: both the exotic’s contracts and the strategy’s already run on simulated paths, by the same fold as production (C-2.8, C-6.1); the marginal cost is only declaring the strategy unit’s terms and naming the trajectory. The compute is not thereby cheap — it is one production valuation per step, the exotic’s per-step pricing cost unchanged (VM-10); the ledger makes the *specification* easy, never the pricing.

(*Simulability, Reproducibility, Auditability; C-2.8, C-6.4, C-7.2, C-10.2, C-12.1, MD-4, MD-11, MD-13.*)

### 3. The Life of a Valuation

One unit walks the whole picture. Follow a single long call, struck at 100 on a stock at 100, three months to expiry, marked at a 30% implied volatility. Its value is 5.98, its delta 0.53, its gamma 0.0265, its decay  $-0.047$  per trading day. Each row of Table 1 is a link; each is proven from the one above.

The market-move link also reads as the gamma-and-theta balance of VM-4: the \$2 move clears this option’s breakeven of \$1.89, so a delta-hedged holder’s gamma gain (0.053) just covers a day’s decay (0.047). Inside the breakeven the day loses; outside it the day earns; and that balance, not the delta, is what a long option position is really holding.

**A shifted-world column.** From the base mark, a risk report is one more valuation under a recorded shift. Shift spot  $+20\%$ , to 120, and re-apply the same recipe. The greek estimate is  $0.53 \times 20 + \frac{1}{2} \times 0.0265 \times 20^2 \approx +15.9$ ; the full revaluation is  $+14.9$  (the mark moves to 20.9). The gap,  $\approx +1.0$  — near a fifteenth of the move, because a twenty-percent shift is far outside the regime where the quadratic is a good control — is not an error to be suppressed but a recorded

| Link        | What changed                                               | Explain into this mark                                                                          |
|-------------|------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Base        | struck at all coordinates                                  | value 5.98; delta 0.53, gamma 0.0265                                                            |
| Carry day   | one trading day, spot flat                                 | carry $-0.047$ ; residual $< 0.001 \Rightarrow 5.93$                                            |
| Market move | spot 100 $\rightarrow$ 102, vol held                       | delta $+1.060$ , gamma $+0.053$ ;<br>ex-ante residual $-0.001$ (bound 0.010) $\Rightarrow 7.09$ |
| CA sandwich | 2:1 split; strike $\rightarrow$ 50,<br>deliverable doubles | frame re-coordination (zero-profit);<br>economics 0; residual $\approx 0 \Rightarrow 7.09$      |

Table 1: A valuation chain of four links on one unit. Each link isolates one effect for legibility; on a real book carry and market move fall in a single link and both appear as explain lines. The market-move residual is the ex-ante gap (the higher-order and gamma-variation content), well inside its declared bound. The corporate-action link is a sandwich: the pre-ex value 7.09 in the old frame equals the post-ex value 7.09 in the new one, and the near-zero residual is the proof the split was applied consistently — not a phantom valuation from a new share count against a stale price (C-11.3).

diagnostic (VM-10). The shift composes: add a volatility-down leg and the same recipe returns a new vector, no second code path.

**A link that breaks.** Now swap the call for a digital that pays a fixed coupon if the stock is above 100 at expiry, and take a link that straddles the strike a week out. The two-point explain fails on sight: the reprice is the whole coupon, while the endpoint deltas, evaluated where the value has already flattened either side of the strike, read near zero, so the entire notional lands in the residual —  $O(1)$ , not  $O(\|\delta\|^4)$ . The certificate does not hide this: the residual breaches its bound, the breach is booked as a named line (VM-6), and the chain is flagged broken until the position is re-marked (VM-7). The honest attribution here is a gap scenario with a probability attached, not a greek — which is exactly what VM-4 warned the smooth explain could not see, made visible by the very residual it refuses to absorb.

**Disputed.** A counterparty contests the collateral mark. It is not argued: replay exhibits the chain, its certificates, and its coordinates, and reproduces the mark bit for bit, the as-at pinned to the mark’s own cut (VM-8, MD-14). Replay cannot decide whose surface or convention is right, but it localises the disagreement to a differing input, frame, model, or convention.

## 4. What This Manifesto Does Not Govern

This manifesto governs how valuation and profit-and-loss are recorded, proven, and projected. It does not govern **which model, surface, prior, or re-marking convention is correct** — model choice, whose test is hedging performance, out of scope (C-Scope.11, MD-9, MD-15); nor **model governance, validation, and retention** — a valuation reproduces from recorded prices unconditionally and from a model’s number only if the model is kept (C-14.15); nor the **numerical methods** by which a model is solved — regression, quadrature, surrogate pricing — which are the reference implementation’s, not the specification’s (C-14.9); nor **price formation**, the **terms of a corporate action** (they arrive as recorded events, MD-13), or **settlement**. These lie outside the Ledger boundary and are reconciled against it only at its perimeter.

## 5. Subordination

This manifesto is subordinate to two parents — the Architectural Constitution and the Market Data Manifesto — and rated against both, never the other way round. Its vocabulary is theirs: one name per component, and no synonym for any fixed term (C-Auth.4). Where a principle here would conflict with either parent, the parent prevails and the conflict is parked with the

exact amendment text and the exact clause it replaces — never resolved by quiet narrowing, which for a guarantee of either parent is the worse failure. A genuine departure amends this document explicitly first; an article identifier VM- $n$  is never reused or renumbered.

## Conclusion

Valuation has the same root cause of failure as everything else the Ledger touches: the same fact held in two places, marked by two logics, drifting apart. The Ledger removes it here as everywhere. A valuation is a projection of the one record; each model price it consumes is an observation; each mark it produces is proven from the last by a profit-and-loss explain whose residual it must keep below a declared bound, and an unexplained move is a broken chain, not a number to carry. Every mark carries its coordinates, is dispute-ready by replay, survives a corporate action as a frame re-coordination that is not profit-and-loss, and is re-run in a simulated world for risk by the very same recipe. Operate it any other way and the divergence the architecture exists to remove comes back; this way, a valuation is a fact, and there is nothing left to reconcile.

## PART B — The Pricing Engine

*One fact about the pricing function the chain stores*

Part A governs the record and its chain: a mark is a re-entered observation (VM-1), proven from the last by a profit-and-loss explain whose residual sits below a declared bound (VM-4, VM-6). Part B is *deliberately distinct*. It is not workflow doctrine; it steps beneath the doctrine to state, and derive, one mathematical fact about the pricing function the chain stores — the fact beneath a single line of the explain, the line that reconciles carry against convexity. That fact: the price and the sensitivities a book hedges with are, in general, drawn from two different mathematical worlds, and the object that measures the gap is a *product instantaneous volatility*. Part B is descriptive of that structure only. It proposes no model, no surface, no re-marking rule, and no remedy beyond the single monitoring mandate of PE-6; the rule by which an implied surface moves enters solely as a declared mathematical map, never as an account of anyone's practice. Its articles are numbered PE- $n$ , append-only, and cite Part A by VM- $n$  and the corpus by section and equation.

**PE-1. The objects: a pricing model, its Feynman–Kac equation, and a declared surface-move map.** PE-1 Fix a filtered probability space carrying a risk-neutral measure  $\mathbb{Q}$  and a strictly positive underlying  $S_t$ . The following are standing for all of Part B.

**(A1) Carry.** A constant short rate  $r \geq 0$  and continuous dividend yield  $q \geq 0$ ; the drift of  $S$  under  $\mathbb{Q}$  is  $(r - q)$ . The corpus's zero-rate specialization  $r = q = 0$ , hence  $F = S$ , is flagged at each use.

**(A2) Pricing model  $\mathfrak{M}$ .** Its diffusion coefficient is a function  $\sigma_{\text{mod}}$ : either *constant*  $\sigma_{\text{mod}} \equiv \sigma > 0$  (Black–Scholes for vanillas), or *local*  $\sigma_{\text{mod}} = \sigma_{\text{loc}}(S, t)$  with  $0 < \sigma_{\min} \leq \sigma_{\text{loc}} \leq \sigma_{\max} < \infty$  and locally Lipschitz in  $S$  uniformly in  $t$  on compacts, so that  $dS_t = (r - q)S_t dt + \sigma_{\text{mod}}(S_t, t) S_t dW_t^{\mathbb{Q}}$  has a unique strong solution.

**(A3) Product and price.** A product  $\Phi$  with model price  $P(S, t) = \mathbb{E}^{\mathbb{Q}}[e^{-r(T-t)}\Phi \mid S_t = S]$ , assumed  $C^{2,1}$  on the open domain  $\mathcal{O} = (0, \infty) \times [0, T)$ . For a payoff with a discontinuity or a kink in the state — a digital, a barrier, an autocallable —  $\mathcal{O}$  excludes the discontinuity locus, and  $C^{2,1}$  fails on it; this is the same non-smoothness Part A books as an  $O(1)$  residual (VM-4), and PE-3 shows where it re-enters here.

**(A4) Implied surface.** An arbitrage-free implied-volatility surface  $\Sigma_t$ , calibrated to market at  $t$  and valid in price space (MD-15); its scalar value at the product's reference strike and expiry is the implied volatility  $\sigma$ .

**(A5) The surface-move map (the hedging rule, as an object).** A declared *spot-dynamic*  $\mathcal{D}$ : a map that assigns to a spot level  $S$  a surface  $\Sigma(\cdot; S)$ , of class  $C^2$  in  $S$  near the valuation spot, so that the reprice map

$$g_{\Phi, \mathcal{D}}(S) := P(S, \Sigma(\cdot; S)) \tag{1}$$

is twice differentiable there. The moved surface stays admissible:  $\Sigma(\cdot; S) \in \mathcal{A}$ , the set of arbitrage-free surfaces (Vol I, §2.6), for  $S$  in the neighbourhood; since  $\mathcal{A}$  is open (Vol I, Def. 2.20), this holds automatically when the base surface is interior to  $\mathcal{A}$  and  $\mathcal{D}$  is continuous into surface space.  $\mathcal{D}$  is a mathematical object only: sticky-strike, sticky-moneyness, and sticky-delta are instances, as is the corpus dynamic — the decomposition  $\sigma = \sigma_{\text{ATM}}(\ell_F, T) + \sigma_{\text{smile}}(m, T)$  (Vol I, §2.1.1, Eq. (2.1)) *held fixed* as the forward  $F = F(S)$  floats (Eqs. (2.19)–(2.20), slope vol.slope) — each a different  $\mathcal{D}$ . The decomposition (2.1) is a coordinate system, not a dynamic; the dynamic is that system carried under the forward map. This map  $\mathcal{D}$  is not a new coordinate: it is the surface-dynamics content of the re-marking convention Part A already declares (VM-5) — one

declared, recorded term (MD-6), read in two roles, the attribution split in Part A and the hybrid Greeks and  $\sigma_{\text{prod}}^2$  here. Accordingly the *shadow gamma* of PE-5 is the second-order instance of VM-5's *shadow greek* — the value routed through the held-invariant factor — not a separate object.

**Claim (Feynman–Kac).** Under (A1)–(A3),  $P$  satisfies on  $\mathcal{O}$

$$\partial_t P + \frac{1}{2} \sigma_{\text{mod}}(S, t)^2 S^2 \partial_{SS} P + (r - q) S \partial_S P - rP = 0, \quad P(S, T) = \Phi. \quad (2)$$

This is the Feynman–Kac PDE of  $\mathfrak{M}$  — the model's own pricing equation, the consistency its price and Greeks satisfy by construction; its diffusion coefficient is  $\sigma_{\text{mod}}$ , the volatility function of the model used to price. Under (A2) the standard Feynman–Kac hypotheses hold (uniform ellipticity and local regularity of the coefficients), so (2) is exact, not a heuristic.

(Correctness; VM-1, VM-2, MD-15; Vol I §1, Assumption 1.1.)

**PE-2. The discrepancy: hybrid Greeks do not satisfy the model's Feynman–Kac equation.** PE-2 The Greeks  $\mathfrak{M}$  owns are the partials  $\partial_S P$ ,  $\partial_{SS} P$ ,  $\partial_t P$  of the same  $P$ ; by construction they combine to zero in (2). The Greeks a book hedges with are, in general, different objects: they are read off the reprice map (1), in which the surface is moved by  $\mathcal{D}$  and the product is repriced.

**Definition (hybrid Greeks).** The *hybrid delta* and *hybrid gamma* are the total derivatives of (1) along the curve  $S \mapsto (S, \Sigma(\cdot; S))$ ,

$$\Delta_{\text{hyb}} := g'_{\Phi, \mathcal{D}}(S) = \partial_S P + \underbrace{D_{\Sigma} P[\partial_S \Sigma]}_{=: \delta_{\Delta}}, \quad (3)$$

$$\Gamma_{\text{hyb}} := g''_{\Phi, \mathcal{D}}(S) = \partial_{SS} P + \underbrace{2 D_{\Sigma}(\partial_S P)[\partial_S \Sigma] + D_{\Sigma \Sigma} P[\partial_S \Sigma, \partial_S \Sigma] + D_{\Sigma} P[\partial_{SS} \Sigma]}_{=: \delta_{\Gamma}}, \quad (4)$$

where  $D_{\Sigma} P[\cdot]$  is the Gâteaux derivative of  $P$  in the surface argument. These directional (Gâteaux) derivatives are taken along the declared  $C^2$  curve  $S \mapsto \Sigma(\cdot; S)$  of (A5); their existence is the standing hypothesis that  $P$  is twice differentiable in the surface argument there — realized concretely by the smooth Black–Scholes dependence of PE-5 — not a claim of Fréchet differentiability on surface space, which the manifesto neither needs nor asserts. Coordinatized by strike,  $\delta_{\Delta} = \int (\partial P / \partial \sigma(K)) (\partial \sigma(K) / \partial S) dK$  is the same object. The corrections  $\delta_{\Delta}, \delta_{\Gamma}$  are the surface-response terms; they are exactly the smile-adjustment terms of the total (routed) Greeks of Vol I, §2.4.2 (Eq. (2.21)), made explicit per instance in PE-5. Both  $\delta_{\Delta}$  and  $\delta_{\Gamma}$  are proportional to the surface response  $\partial_S \Sigma$  that  $\mathcal{D}$  declares.

**Claim.** With the model theta  $\Theta := \partial_t P$  and the hybrid Greeks  $\Delta_{\text{hyb}}, \Gamma_{\text{hyb}}$ , the Feynman–Kac operator of (2) does not vanish:

$$\mathcal{L}_{\text{hyb}} := \Theta + \frac{1}{2} \sigma_{\text{mod}}^2 S^2 \Gamma_{\text{hyb}} + (r - q) S \Delta_{\text{hyb}} - rP = \frac{1}{2} \sigma_{\text{mod}}^2 S^2 \delta_{\Gamma} + (r - q) S \delta_{\Delta}, \quad (5)$$

which is nonzero whenever the surface response is nonzero.

**Derivation.** Substitute (3)–(4) into  $\mathcal{L}_{\text{hyb}}$  and add and subtract the model partials:

$$\begin{aligned} \mathcal{L}_{\text{hyb}} &= \underbrace{\left[ \partial_t P + \frac{1}{2} \sigma_{\text{mod}}^2 S^2 \partial_{SS} P + (r - q) S \partial_S P - rP \right]}_{=0 \text{ by (2)}} \\ &\quad + \frac{1}{2} \sigma_{\text{mod}}^2 S^2 (\Gamma_{\text{hyb}} - \partial_{SS} P) + (r - q) S (\Delta_{\text{hyb}} - \partial_S P). \end{aligned}$$

The bracket is the Feynman–Kac equation, identically zero; the remaining two terms are  $\frac{1}{2} \sigma_{\text{mod}}^2 S^2 \delta_{\Gamma} + (r - q) S \delta_{\Delta}$ , giving (5). The discrepancy  $\mathcal{L}_{\text{hyb}} = 0$  exactly when the surface-response

corrections  $\delta_\Delta, \delta_\Gamma$  combine to zero. Sufficient are a surface-insensitive product ( $D_\Sigma P = 0$ , e.g. a forward) or a rule that holds the surface still *to second order* ( $\partial_S \Sigma = \partial_{SS} \Sigma = 0$ ) — note  $\delta_\Gamma$  retains the term  $D_\Sigma P[\partial_{SS} \Sigma]$ , so  $\partial_S \Sigma = 0$  alone does not suffice. The exact condition is that  $\mathcal{D}$  reproduce the model's own endogenous surface dynamics (Remark below).  $\square$

**Remark (two worlds, and when they coincide).**  $P$  is the price under a diffusion whose coefficient is  $\sigma_{\text{mod}}$ ; in that world the implied surface is an *output*, and its response to spot is whatever the model dictates. The hybrid Greeks encode an *externally declared* response  $\partial_S \Sigma$ . Equation (5) is precisely the mismatch between the two: it vanishes exactly when  $\mathcal{D}$  reproduces the model's own endogenous surface dynamics, and is otherwise the signed amount by which the hedging Greeks fail the model's own equation. The pair  $(P, \partial P_{\text{model}})$  solves (2); the pair  $(P, \partial P_{\text{hyb}})$  does not — same price, sensitivities from a different world.

(Correctness; VM-4, VM-5; Vol I §2.4.2, Eq. (2.21), §3.6, Eq. (3.34).)

**PE-3. Product instantaneous volatility: the coefficient that restores the equation, and why it need not be smooth.** PE-3 The discrepancy (5) can be absorbed into a single coefficient. Since  $\sigma_{\text{mod}}$  enters (2) only through  $\frac{1}{2}\sigma_{\text{mod}}^2 S^2 \partial_{SS} P$ , one asks for the diffusion coefficient that makes a PDE of the same form hold with the model theta and the hybrid delta and gamma.

**Definition (product instantaneous variance and volatility).** At a point  $(S, t)$  with  $\Gamma_{\text{hyb}}(S, t) \neq 0$ , the *product instantaneous variance*  $\sigma_{\text{prod}}^2(S, t)$  is the sign-carrying number solving

$$\partial_t P + \frac{1}{2} \sigma_{\text{prod}}^2 S^2 \Gamma_{\text{hyb}} + (r - q) S \Delta_{\text{hyb}} - rP = 0, \quad (6)$$

that is, in closed form,

$$\sigma_{\text{prod}}^2 = - \frac{2[\Theta + (r - q) S \Delta_{\text{hyb}} - rP]}{S^2 \Gamma_{\text{hyb}}}, \quad \Theta = \partial_t P. \quad (7)$$

It is the first-class object here — real and total wherever  $\Gamma_{\text{hyb}} \neq 0$ , its sign a diagnostic in its own right; the *product instantaneous volatility*  $\sigma_{\text{prod}} = \sqrt{\sigma_{\text{prod}}^2}$  is its nonnegative root where  $\sigma_{\text{prod}}^2 \geq 0$ . Under  $r = q = 0$  this is  $\sigma_{\text{prod}}^2 = -2\Theta/(S^2 \Gamma_{\text{hyb}})$ ; with  $q = 0$  it matches the normative form  $\sigma_{\text{prod}}^2 = -2(\theta + rS\Delta_{\text{hyb}} - rP)/(S^2 \Gamma_{\text{hyb}})$ .

**Well-posedness (stated, not waved).** Equation (7) defines  $\sigma_{\text{prod}}^2$  wherever  $\Gamma_{\text{hyb}} \neq 0$ . Writing  $N := -2[\Theta + (r - q)S\Delta_{\text{hyb}} - rP]$  and  $M := S^2 \Gamma_{\text{hyb}}$ , the value  $\sigma_{\text{prod}} = \sqrt{N/M}$  is real and nonnegative iff  $N$  and  $M$  share sign; where  $M = 0$  the ratio has a pole and  $\sigma_{\text{prod}}$  is undefined; where  $N/M < 0$  no real  $\sigma_{\text{prod}}$  exists — the FK form cannot be restored with a real coefficient; and where  $\mathcal{D}$  carries  $\Sigma(\cdot; S)$  out of the admissible set  $\mathcal{A}$  (PE-1, A5), the model price of a surface-dependent product is itself undefined, and  $\sigma_{\text{prod}}$  inherits that gap. These are structural facts about the hybrid pair, recorded, not smoothed.

**Its nature, stated with total honesty.** (i) *Reverse-engineered, not an SDE coefficient.*  $\sigma_{\text{prod}}$  is defined by algebraically inverting (6) given  $(\Theta, \Delta_{\text{hyb}}, \Gamma_{\text{hyb}})$ ; it is not the diffusion coefficient of any stochastic differential equation for  $S$ . In particular  $\sigma_{\text{prod}} \neq \sigma_{\text{mod}}$ : it is neither the constant  $\sigma$  nor  $\sigma_{\text{loc}}$ . (ii) *Specific to the product and to the rule.* Through  $\Delta_{\text{hyb}}, \Gamma_{\text{hyb}}$  the right-hand side of (7) depends on the product  $\Phi$  (via its surface sensitivities) and on  $\mathcal{D}$  (via  $\partial_S \Sigma, \partial_{SS} \Sigma$ ). Two products at one  $(S, t)$  under one rule have different  $\sigma_{\text{prod}}$ ; one product under two rules (say sticky-strike versus sticky-delta) has different  $\sigma_{\text{prod}}$ . (iii) *Fitted, hence not required to be smooth* — and here is why.  $\sigma_{\text{prod}}^2$  is the ratio  $N/M$  of two independently constructed fields; it solves no differential equation that would impose regularity, so its only regularity is that inherited from the quotient. The denominator  $M = S^2 \Gamma_{\text{hyb}}$  can vanish and change sign — Vol I exhibits  $\Gamma_{\text{adj}}$  crossing zero as Vanna does (Vol I, §3.9.3) — while the numerator  $N$  need not, so  $\sigma_{\text{prod}}^2$

carries poles and sign changes there. Even where finite,  $N$  and  $M$  inherit  $\partial_S \Sigma$  and  $\partial_{SS} \Sigma$  from the declared rule  $\mathcal{D}$ , itself only a declared map — admissibly kinked (a sticky rule piecewise in moneyiness) — so nothing forces  $\sigma_{\text{prod}}$  to be even continuous. Contrast the Dupire local volatility, which is a smooth functional of a smooth arbitrage-free surface (Vol I, §2.5):  $\sigma_{\text{loc}}$  is *modelled*;  $\sigma_{\text{prod}}$  is *fitted*, and the two regularities are not the same.

(Correctness; VM-4, VM-6; Vol I §2.5, §3.9.)

**PE-4. Carry: theta is model-determined; the theta–gamma balance closes only against the product instantaneous volatility.** PE-4

**Claim 1 (theta is invariant to the surface-vs-spot dynamic).** The model theta  $\Theta = \partial_t P$  does not depend on the choice of  $\mathcal{D}$ .

**Derivation.**  $\Theta$  is the partial of the model price in calendar time at *fixed* spot. By (A5)  $\mathcal{D}$  is a spot-dynamic: it prescribes the surface response  $\partial_S \Sigma$  and enters the reprice map only through  $S$ -derivatives, where the chain rule multiplies the surface terms  $D_\Sigma P[\cdot]$  by  $\partial_S \Sigma$  — see (3)–(4). The calendar partial at fixed  $S$  contains no  $\partial_S \Sigma$  factor; hence  $\Theta$  is one and the same across the whole admissible family of surface-move maps, while  $\Delta_{\text{hyb}}, \Gamma_{\text{hyb}}$  vary with  $\mathcal{D}$  through  $\delta_\Delta, \delta_\Gamma$ . This is the asymmetry the normative source states: the dynamics of the implied surface with respect to spot affect the hybrid delta and gamma and leave theta untouched.  $\square$

**Scope (honest caveat).** The invariance holds because  $\mathcal{D}$  is declared as a spot-dynamic. A rule that additionally prescribed an autonomous calendar roll  $\partial_t \Sigma$  independent of the model would give a “hybrid theta” carrying  $D_\Sigma P[\partial_t \Sigma]$ ; PE-1(A5) fixes  $\mathcal{D}$  as a spot-dynamic precisely so that theta is the model’s alone.

**Claim 2 (the theta–gamma balance is readable only at  $\sigma_{\text{prod}}$ ).** Rearranging (6),

$$\Theta + (r - q)S \Delta_{\text{hyb}} - rP = -\frac{1}{2} \sigma_{\text{prod}}^2 S^2 \Gamma_{\text{hyb}}, \quad (8)$$

and under the zero-rate convention ( $r = q = 0$ ,  $F = S$ , flagged),  $\Theta = -\frac{1}{2} \sigma_{\text{prod}}^2 S^2 \Gamma_{\text{hyb}}$ . The hedger carries  $\Gamma_{\text{hyb}}$ ; the daily breakeven move — the  $|\delta S|$  at which gamma gain offsets a day’s decay — follows from  $\frac{1}{2} \Gamma_{\text{hyb}} (\delta S_{\text{BE}})^2 = |\Theta| dt$  and (8) as

$$\delta S_{\text{BE}} = \sigma_{\text{prod}} S \sqrt{dt}, \quad (9)$$

the same form as the corpus breakeven  $\sigma S / \sqrt{252}$  (Vol I, §1.4, Eq. (1.13)) but with  $\sigma_{\text{prod}}$  in place of the implied  $\sigma$  — and it presumes the real, long-carry regime  $\sigma_{\text{prod}}^2 > 0$ ; where PE-3 gives  $\sigma_{\text{prod}}^2 \leq 0$  (a pole, or a sign flip) there is no real breakeven. So  $\sigma_{\text{prod}}$  is, by construction, the one volatility at which the model theta and the hedger’s gamma reconcile: a volatility  $\sigma$  leaves in (8) the residual  $\frac{1}{2}(\sigma^2 - \sigma_{\text{prod}}^2) S^2 \Gamma_{\text{hyb}}$ , nonzero for every  $\sigma \neq \sigma_{\text{prod}}$  and equal to the Feynman–Kac discrepancy  $\mathcal{L}_{\text{hyb}}$  of (5) exactly at the model’s own  $\sigma_{\text{mod}}$ .

**Linkage to Part A (the statement threaded there).** The profit-and-loss explain certificate’s carry line reconciles the model theta against the gamma the book actually carries — the hybrid gamma. That reconciliation holds at exactly one volatility,  $\sigma_{\text{prod}}$ . A certificate that reconciles at a volatility other than  $\sigma_{\text{prod}}$  carries a nonzero residual into  $R$  — the model’s  $\sigma_{\text{mod}}$  leaves exactly  $\mathcal{L}_{\text{hyb}} dt$  of (5), a general  $\sigma$  leaves  $\frac{1}{2}(\sigma^2 - \sigma_{\text{prod}}^2) S^2 \Gamma_{\text{hyb}} dt$  — which is exactly the unexplained profit-and-loss VM-6 forbids to leave silently unattributed and VM-7 calls a broken chain. Hence the carry line carries a declared, recorded term naming which volatility reconciles it; VM-4 and VM-6 carry that thread.

(Correctness, Auditability; VM-4, VM-6, VM-7; Vol I §1.4, Eq. (1.13).)

**PE-5. The corpus illustration: the total gamma under an implied-volatility dynamic** PE-5

**drives the product instantaneous volatility, vanillas included.** Take a single name, the constant-vol model priced at the option's implied volatility, and the corpus surface-move map. All under the zero-rate convention  $F = S$ , flagged.

**The declared dynamic, explicitly.** Write the implied vol at the option's fixed strike through the corpus decomposition (Vol I, §2.1.1, Eq. (2.1))  $\sigma(m, \ell_F, T) = \sigma_{\text{ATM}}(\ell_F, T) + \sigma_{\text{smile}}(m, T)$ , with  $m = \ln(K/F)$ ,  $\ell_F = \ln(F/F_{\text{ref}})$ , skew  $s(m) = \partial_m \sigma_{\text{smile}}$ , curvature  $c(m) = \partial_m^2 \sigma_{\text{smile}}$ , and ATM slope  $\text{vol.slope} = \partial_{\ell_F} \sigma_{\text{ATM}}$ . The map  $\mathcal{D}$  is carried entirely by (vol.slope,  $s$ ,  $c$ ): routing through  $F(S)$  gives (Vol I, §2.4.2, Eqs. (2.19)–(2.20))

$$\frac{d\sigma}{dS} = \frac{\text{vol.slope} - s(m)}{S}, \quad \frac{d^2\sigma}{dS^2} = \frac{(\partial_{\ell_F}^2 \sigma_{\text{ATM}} - \text{vol.slope}) + (c(m) + s(m))}{S^2}. \quad (10)$$

**The hybrid gamma, every cross term explicit.** The price is  $C(S, \sigma(S))$  with  $\sigma(S)$  read off the surface under  $\mathcal{D}$ . Its second total spot derivative is the total (smile-adjusted) gamma of Vol I, §2.4.2, Eq. (2.21):

$$\Gamma_{\text{hyb}} = \Gamma_{\text{adj}} = \Gamma + 2 \text{Vanna} \frac{d\sigma}{dS} + \text{Volga} \left( \frac{d\sigma}{dS} \right)^2 + \nu \frac{d^2\sigma}{dS^2}, \quad (11)$$

with  $\Gamma = \partial_{SS} C$ ,  $\text{Vanna} = \partial_S \partial_\sigma C$ ,  $\text{Volga} = \partial_\sigma^2 C$ ,  $\nu = \partial_\sigma C$  the frozen-vol Black–Scholes Greeks. This is exactly the  $\delta_\Gamma$  of (4) in the single-implied-vol realization: the model gamma  $\Gamma$ , plus the vanna and volga terms in the surface velocity  $d\sigma/dS$ , plus vega times the surface acceleration  $d^2\sigma/dS^2$ . It is confirmed to be the object that enters the P&L identity in Vol I, §3.6, Eq. (3.34).

**The product instantaneous volatility, exhibited.** The constant-vol model theta is  $\Theta = \partial_t C = -\frac{1}{2} \sigma^2 S^2 \Gamma$  (Feynman–Kac (2) at  $r = q = 0$  with  $\sigma_{\text{mod}} = \sigma$ ,  $\partial_{SS} C = \Gamma$ ). Insert into (7) with  $\Gamma_{\text{hyb}} = \Gamma_{\text{adj}}$ :

$$\sigma_{\text{prod}}^2 = \frac{-2\Theta}{S^2 \Gamma_{\text{adj}}} = \sigma^2 \frac{\Gamma}{\Gamma_{\text{adj}}}, \quad \text{i.e.} \quad \sigma_{\text{prod}} = \sigma \sqrt{\frac{\Gamma}{\Gamma_{\text{adj}}}}. \quad (12)$$

The model theta fixes the numerator and is invariant to  $\mathcal{D}$  (PE-4); the entire dependence on the assumed dynamic sits in the ratio  $\Gamma/\Gamma_{\text{adj}}$ , and by (11) that ratio departs from 1 for *any* product carrying vega, vanna, or volga once the surface moves with spot ( $\text{vol.slope} \neq 0$  or  $s(m) \neq 0$ ). A vanilla carries all three; therefore  $\sigma_{\text{prod}} \neq \sigma$  for a vanilla under any nontrivial surface dynamic — the normative source's “vanilla options included” is exactly (12). The clean ratio is specific to the constant-vol model: for a local-vol model  $\sigma_{\text{prod}}$  is the general form (7) with  $\Theta = -\frac{1}{2} \sigma_{\text{loc}}^2 S^2 \partial_{SS} P$  and  $\partial_{SS} P$  the local-vol price's own second partial — the  $\sqrt{\Gamma/\Gamma_{\text{adj}}}$  collapse does not survive.

**A number.** Take Part A's base call ( $\sigma = 30\%$ ,  $\Gamma = 0.0265$ , vega  $\approx 0.20$  per vol point,  $S = 100$ ,  $T = 0.25$ ) under a downward skew of one vol point per one-percent up-move ( $d\sigma/dS \approx -0.01$ ). Its vanna is  $\approx 0.10$ , so the surface-velocity term  $2 \text{Vanna} (d\sigma/dS) \approx -0.002$  lowers the effective gamma to  $\Gamma_{\text{adj}} \approx 0.0245$ , and  $\sigma_{\text{prod}} = \sigma \sqrt{\Gamma/\Gamma_{\text{adj}}} \approx 30\% \times \sqrt{0.0265/0.0245} \approx 31\%$ . The breakeven volatility the delta-hedged holder actually runs is about 31%, not the 30% quoted — a vol point of carry the implied number never showed.

**Non-smoothness, exhibited.** Vol I, §3.9.3, exhibits  $\Gamma_{\text{adj}}$  running from a positive smile adjustment at one spot to a negative one a finite move away, as Vanna crosses zero. Where  $\Gamma_{\text{adj}} \rightarrow 0$ , (12) gives  $\sigma_{\text{prod}} \rightarrow \infty$  (a pole); where  $\Gamma_{\text{adj}}$  takes the sign opposite to  $\Gamma$ ,  $\sigma_{\text{prod}}^2 < 0$  and no real  $\sigma_{\text{prod}}$  exists. So even for a vanilla,  $\sigma_{\text{prod}}$  is neither globally smooth nor globally real — the abstract statement of PE-3(iii) made concrete.

**The multi-name instance.** When the book's value depends on an implied correlation re-marked by a declared convention, the surface-response term  $\delta_\Gamma$  of (4) acquires a correlation channel:

the *shadow gamma* of Vol II, §2.5, Eq. (2.23) — under the equal-vol, locally-constant-vol.slope hypothesis of its Prop. 2.27 —

$$\text{ShadowGamma} = \frac{\text{vol.slope}^2}{S} \left[ \partial_\rho^2 V R^2 + \partial_\rho V \partial_\sigma^2 \rho|_\xi \right] + \text{vol.slope} R \left( \partial_{S\rho}^2 V + \frac{\text{vol.slope}}{S} \partial_\sigma^2 V \right),$$

$$R = \partial_\sigma \rho|_\xi,$$

under the sticky-CVC convention ( $d\xi = 0$ ), and vanishing identically under sticky- $\rho$  ( $d\rho = 0$ ). It enters  $\Gamma_{\text{hyb}}$  on its raw footing  $\Gamma^{\text{sh}}$  (Vol II, §2.5, Eq. (2.21)), so that here too the choice of surface-move convention drives  $\sigma_{\text{prod}}$  through (7) — the same structure as the single-name case, with Cega and the correlation dynamics in place of vega and the vol dynamics.

(*Correctness*; VM-4, VM-5; Vol I §2.4.2, Eqs. (2.19)–(2.21), §3.9.3; Vol II §2.5, Eqs. (2.21), (2.23), Prop. 2.27.)

**PE-6. The product instantaneous variance is a first-class derived quantity: computed on demand per product and per valuation over the chain.** PE-6 The canonical object is the *product instantaneous variance*  $\sigma_{\text{prod}}^2$  of (7) — the sign-carrying effective variance, real and total wherever  $\Gamma_{\text{hyb}} \neq 0$  — with the product instantaneous volatility  $\sigma_{\text{prod}} = \sqrt{\sigma_{\text{prod}}^2}$  its nonnegative root where  $\sigma_{\text{prod}}^2 \geq 0$ . It is a *projection* of the recorded Greeks: the deterministic algebraic function (7) of  $\Theta, \Delta_{\text{hyb}}, \Gamma_{\text{hyb}}, P$  and the market state, so it reproduces *unconditionally* from the record with no model re-run — stronger than a mark, whose number needs its retained model (C-14.15). Being a projection it is computed when needed and never stored (C-4.11, MD-6): for every product and every valuation its value, its sign, its poles, and its regularity are monitored by *recomputation over the chain's recorded Greeks*, which is authoritative, so it cannot drift (VM-8). Where the object is pathological the pathology is what the recomputation returns: at a pole ( $\Gamma_{\text{hyb}} = 0$ )  $\sigma_{\text{prod}}^2$  is undefined and the fact is the pole itself; where  $\sigma_{\text{prod}}^2 < 0$  the fact is that no real volatility restores the Feynman–Kac form — the sign is the diagnostic, not an error to be smoothed. Its evolution across links is thereby auditable and reproducible under challenge, an internal diagnostic feeding the certificate of a dispute-ready mark (VM-8), not itself a counterparty mark. *Why the mandate*: a pole or a sign flip in  $\sigma_{\text{prod}}^2$  marks *degenerate carry* — the spot at which the carry line's theta-gamma reconciliation breaks down, the carry-line analogue of the  $O(1)$  residual VM-4 books at a digital. This is the single monitoring mandate of Part B; it prescribes nothing further. *Parked*: the mandate that  $\sigma_{\text{prod}}^2$  additionally be *stored* per valuation on the chain is parked against C-4.11 and MD-6 — a projection stores nothing — and stands in the open-problems index for the owner's ruling; nothing above depends on storage.

(*Auditability, Reproducibility*; VM-1, VM-3, VM-6, VM-8, C-4.11, C-14.15, MD-6, MD-15.)

## Open Problems (Parked Items)

The framework parks a genuine conflict rather than fudge it: it is recorded here with the exact clause it would replace, the exact amendment text, and the justification, and the decision is the owner's alone. This index holds the document's first entry.

### PARK-1 — storing the product instantaneous variance versus C-4.11.

*Conflict*. Part B mandates that  $\sigma_{\text{prod}}^2$  (PE-6) be *stored* per product and per valuation on the chain. This conflicts with the Constitution's C-4.11 and with MD-6:  $\sigma_{\text{prod}}^2$  is a *projection* — an algebraic recomputation over recorded Greeks, needing no model run (PE-6) — and a projection stores nothing. It would be the architecture's first stored ledger-computable output. The in-force text of PE-6 therefore runs the conforming reading — computed on demand, monitored by recomputation, never stored — and only the storage half of the mandate is parked.

*Exact clause it would replace (C-4.11, final sentence).* “Anything computable from the coordinates is a projection, computed when needed and never stored.”

*Exact proposed amendment text.* “Anything computable from the coordinates is a projection, computed when needed; recomputation from the record is always authoritative. A projection’s value may be recorded as a verifiable field of a re-entered observation or a certificate — a **materialised projection** — only where (i) recomputation from the record remains authoritative and always available, (ii) the recorded value carries complete lineage to the inputs it was computed from, and (iii) any correction to those inputs flags the recorded value stale. So recorded, it is never a second source that can silently diverge, and recomputation is the standing check. Absent all three, a projection is never stored.” Mirrored in MD-6: “stores nothing” becomes “stores nothing, save as a materialised projection under C-4.11’s three conditions.”

*Justification.* A recomputable, lineage-bearing, flagged-stale field cannot silently diverge, so the reconciliation-failure mode the architecture exists to remove (C-1.2, C-1.3) stays closed and a materialised  $\sigma_{\text{prod}}^2$  series would be operationally safe. Safety is not textual permission: C-4.11 is categorical, and amending it is the owner’s alone.

*Alternative resolution, keeping C-4.11 literal.* The storage mandate’s “stored” becomes “computed on demand and monitored,”  $\sigma_{\text{prod}}^2$  a projection never stored — exactly the conforming reading PE-6 already runs; under it no amendment is needed and the park closes.